

Wajdi J. Alnoush

alnoushw@mcmaster.ca |    | Hamilton, ON

Education

Doctor of Philosophy, Ph.D.

McMaster University

Chemical Engineering
2026

Master of Science, M.S.

Texas A&M University

Chemical Engineering
2018

Bachelor of Science, BSc.

Texas A&M University

Chemical Engineering
2016

Research Experience



McMaster University | 2020 - 2025

Supervisor: [Dr. Drew Higgins](#)

Research: Development and Characterization of M-N-C Electrocatalysts for the Oxygen Reduction Reaction (ORR) and the Carbon Dioxide Reduction Reaction (CO₂RR)



Alliance for AI-Accelerated Materials Discovery, University of Toronto | 2020 - 2024

PIs: [Dr. Drew Higgins](#), [Dr. Cathy Chen](#), [Dr. Dave Sinton](#); Lead PI: [Dr. Ted Sargent](#)

Research: *Operando* X-ray Absorption Spectroscopy (XAS) Characterization of Electro- and Thermo-catalysts for Oxygen Evolution, Carbon Dioxide Reduction and Ethylene Oxidation Reactions.



McMaster-TotalEnergies Laboratory Operando TEM and XAS (LoTX) | 2022 – 2024

Supervisor: [Dr. Drew Higgins](#) and [Dr. Amitava Sarkar](#)

Research: Technique development and characterization of Carbon Dioxide reduction electrocatalysts via Lab-scale *operando* X-ray Absorption Spectroscopy (XAS)



Texas A&M University at Qatar | 2018 - 2019

Supervisor: [Dr. Nayef Alyafei](#)

Research: Wettability Alteration in Carbonate Rocks for Enhanced Oil Recovery (EOR)



Texas A&M University at Qatar | 2016 - 2018

Supervisor: [Prof. Marcelo Castier](#)

Research: Modeling of Natural Gas Supersonic Separation Process



Texas A&M University at Qatar | 2014 - 2016

Supervisor: [Prof. Nimir Elbashir](#)

Research: Enhancement and Optimization of Gas-to-Liquid (GTL) - Derived Synthetic Diesel Properties

Work Experience



Visiting Research Scientist | CERT Systems Inc.

September 2023 – April 2024

- Design and synthesize CO₂RR electrocatalysts selective to multi-carbon products.
- Develop novel CO₂RR cathodes for vapor-fed MEA zero-gap electrolyzers.
- Develop non-PGM OER anodes for CO₂ MEA zero-gap electrolyzers.
- Innovate CO₂ electrolysis strategies to increase energy efficiency at industrially-relevant current densities.
- Analyze electrolyzer and electrode dynamics through *operando* voltage diagnostics and electrochemical impedance spectroscopy (EIS).
- Design and build a pilot-scale automated test platform for MEA electrolyzers.
- Establish an in-house contact-angle/wettability measurement and analysis setup for surface characterization.



Research Associate | Texas A&M University at Qatar

October 2018 – August 2019

Department: **Petroleum Engineering**

- Conduct experimental research to investigate wettability alteration in low salinity water flooding as a secondary means for enhanced oil recovery (EOR).
- Assess surface wettability using contact angle goniometer and surface roughness with atomic force microscopy (AFM)
- Establishment of quality and safety management systems for the multiphase flow in porous media (MFPM) laboratory



Teaching Fellow | Teach for Qatar

June 2016 – July 2018

- Enhance education in Qatar through teaching and research.
- Create and monitor educational outcomes research in collaboration with teach for all.
- Introduce creative technological tools in education such as interactive tools and 360° immersive reality class streaming for parents.
- Lead school STEM and scientific research projects for two consecutive years working on the following projects: "Effect of using GTL diesel as an alternative to conventional diesel in Qatari school buses" and "Using drones as mobile radars in Doha"



Intern | Dolphin Energy

June 2015 – August 2015

Department: **Laboratory**

- Conduct daily laboratory tasks, including Gas Chromatography analysis of product samples, water quality experiments and Sulphur tests.
- Complete technical process training at the laboratory and the operations departments.
- Troubleshooting on facility equipment.



Intern | June 2014 – September 2014

Department: **Operations**

- Complete a series of technical operation training sessions. Training rotations included Health, Safety and Environment (HSE), process management, waste management, plant control room, and maintenance.
- Assist department heads of the low-density polyethylene (LDPE) and linear-low density polyethylene (LLDPE) production units



Intern | June 2013 – August 2013

Department: **Power Plant**

- Shadow power plant operation personnel in operating a combined cycle power plant.
- Assist power plant chemists in field monitoring of cooling tower, heat recovery steam generator, ion-exchange columns, and condenser dosing system.
- Conduct laboratory wastewater analysis and quality tests.
- Assist in safety regulations compliance checks.

Teaching Experience

Teaching Assistant, Chemical Engineering | McMaster University

Numerical Methods and Computing for Chemical Engineers (CHEM ENG 2E04) | Fall 2024

Chemical Engineering Thermodynamics (CHEM ENG 3D03) | Fall 2022

Numerical Methods and Computing for Chemical Engineers (CHEM ENG 2E04) | Fall 2021

Numerical Methods and Computing for Chemical Engineer (CHEM ENG 2E04) | Fall 2020

Teaching Assistant, Petroleum Engineering | Texas A&M University at Qatar

Reservoir Fluids - PETE 310 | Fall 2019

Reservoir Petrophysics - PETE 311 | Spring 2019

Published

1. **Alnough, W.;** Noor, N.; Angizi, S.; Rakhsha, A.; Baker, T.; Higgins, D., Defective Ceria as a Structural and Electronic Modifier of Nitrogen-doped Carbon and Cobalt-Nitrogen-Carbon Electrocatalysts for Oxygen Reduction in Alkaline Electrolyte. *Journal of Materials Chemistry A* **2026**.
2. Rakhsha, A.; Irannezhad, A.; Noor, N.; Suichies, A.; Eraky, H.; **Alnough, W.;** Eslami, R.; Bernar, J.; Ngunjiri, R.; Black, R.; Gilavan, M. T.; Soleymani, L.; Higgins, D., Laser-engraved Graphitized Kapton Electrodes facilitate In situ X-ray Absorption Spectroscopy of Electrocatalysts. *Cell Reports Physical Science* **2026**, 7(3).
3. **Alnough, W.;** Noor, N.; Abdellah, A.; Tan, S.; Angizi, S.; Higgins, D., Mitigating Cobalt Nanoparticles in Pyrolyzed Co-ZIF-Derived Oxygen Reduction Reaction Electrocatalysts in Alkaline Media. *Applied Catalysis B: Environment and Energy* **2025**, 126121.
4. Ismail, F.; **Alnough, W.;** Abdellah, A.; Tan, S.; Salem, K.; Rakhsha, A.; Noor, N.; Fefer, M.; Terazono, Y.; Chen, N.; Higgins, D., Boosting electrochemical conversion of CO₂ to CO in a Membrane Electrode Assembly via Metal-Organic-Framework Derived Nickel–Nitrogen/Carbon Supported Nickel–Zinc Carbide Particle Catalyst. *ACS Electrochemistry* **2025**, 1(7), 1141–1153.
5. Abed, J.; Grigioni, I.; Kose, T.; **Alnough, W.;** Park, S.; Polo, A.; Lee, B.H.; Sinton, D.; Higgins, D.; Sargent, E. H, Zero-Gap Electrolyzers Accelerate Reconstruction of Cu₂O-Derived Catalysts under CO₂ Reduction. *ACS Energy Letters* **2024**, 9, 6225–6232.
6. Abed, J.; Heras-Domingo, J.; Sanspeur, R.; Luo, M.; **Alnough, W.;** Meira, D.; Wang, H.; Wang, J.; Zhou, J.; Zhou, D.; Fatih, K.; Kitchin, J.; Higgins, D.; Ulissi, Z.; Sargent, E., Pourbaix Machine Learning Framework Identifies Acidic Water Oxidation Catalysts Exhibiting Suppressed Ruthenium Dissolution. *Journal of the American Chemical Society* **2024**, 146(23), 15740–15750.
7. e Silva, R.; Dias, E.; Escalona-Durán, F.; **Alnough, W.;** de Oliveira, J.; Higgins, D.; Ribeiro, C., Unveiling BiVO₄ photoelectrocatalytic potential for CO₂ reduction at ambient temperature. *Materials Advances* **2024**, 5(11), 4857–4864.
8. Jahoushi, K.; Ayesh, A.; El-Maghraby, H.; **Alnough, W.;** Higgins, D.; Hassan, F.; Greish, Y., Tunable Hydroxyapatite/Magnetite Nanohybrids with Preserved Magnetic Properties. *Advanced Materials Interfaces* **2022**, 9(16), 2102120.
9. Yang, X.; Zeng, Y.; **Alnough, W.;** Hou, Y.; Higgins, D.; Wu, G., Tuning Two-Electron Oxygen-Reduction Pathways for H₂O₂ Electrosynthesis via Engineering Atomically Dispersed Single Metal Site Catalysts. *Advanced Materials* **2022**, 34(23), 2107954.
10. **Alnough, W.;** Shaat, A.; Alyafei, N. In *What is the Optimum Wettability for Oil Recovery Through Waterflooding?*, SPE Conference at Oman Petroleum & Energy Show, OnePetro: **2022**.
11. Ismail, F.; Abdellah, A.; Lee, H.-J.; Sudheeshkumar, V.; **Alnough, W.;** Higgins, D. C., Impact of Nickel Content on the Structure and Electrochemical CO₂ Reduction Performance of Nickel–Nitrogen–Carbon Catalysts Derived from Zeolitic Imidazolate Frameworks. *ACS Applied Energy Materials* **2022**, 5(1), 430–439.
12. **Alnough, W.;** Black, R.; Higgins, D., Judicious selection, validation, and use of reference electrodes for in situ and operando electrocatalysis studies. *Chem Catalysis* **2021**, 1(5), 997–1013.
13. **Alnough, W.;** Sayed, A.; Solling, T. I.; Alyafei, N., Impact of calcite surface roughness in wettability assessment: Interferometry and atomic force microscopy analysis. *Journal of Petroleum Science and Engineering* **2021**, 203, 108679.
14. **Alnough, W.;** Castier, M., Shortcut modeling of natural gas supersonic separation. *Journal of Natural Gas Science and Engineering* **2019**, 65, 284–300.
15. **Alnough, W.;** Sayed, A.; Alyafei, N., Optimization of contact angle and interfacial tension measurements for fluid/rock systems at ambient conditions. *MethodsX* **2019**, 6, 1706–1715.
16. Abdulhussein, T.; Elhafyan, E.; **Alnough, W.;** Alyafei, N. *Numerical Investigation on the Effect of Wettability and Heterogeneity on Viscous Fingering and Oil Recovery*, 81st EAGE Conference and Exhibition 2019, European Association of Geoscientists & Engineers: **2019**; pp 1–5.
17. Abd, A. S.; Elhafyan, E.; Siddiqui, A. R.; **Alnough, W.;** Blunt, M. J.; Alyafei, N., A review of the phenomenon of counter-current spontaneous imbibition: Analysis and data interpretation. *Journal of Petroleum Science and Engineering* **2019**, 180, 456–470.
18. Selam, M.; **Alnough, W.;** Bohra, M.; Mohamed, H.; Hussain, R.; Elbashir, N. *Role of aromatics in synthetic fuels*, Qatar Foundation Annual Research Conference Proceedings Volume 2014 Issue 1, Hamad bin Khalifa University Press (HBKU Press): **2014**; p EESP0487.

Submitted

1. Abdellah, A.; Salem, K.; Mousavi, S.; **Alnoush, W.**; Teimouri, Z.; Raksha, A.; Ebrahimi, A.; Noor, N.; Eslami, R.; Ismail, F.; Sarkar, A.; Black, R.; Higgins, D., *In situ* Transmission Electron Microscopy of the Structural Evolution and Stability of Cu-Ag Bimetallic Catalysts during the Electrochemical CO₂ Reduction. *In review*
2. Tan, S.; Zhu, M.; Angizi, S.; Yang, X.; **Alnoush, W.**; Park, J.; Hamada, N.; Salem, K.; Zhang, J.; Higgins, D., Integrating High Nickel Loading and Template-Derived Macropores in Nickel-Nitrogen-Carbon Catalysts for High-Performance CO₂ Electrolysis. *In review*

Book Chapters

1. Gas-to-liquid (GTL)-Derived Synthetic Fuels: Role of Additives in GTL-Derived Diesel Fuels; The Water-Food-Energy Nexus: Processes, Technologies, and Challenges; CRCPress **2017**; Chapter 33. [ISBN 9781138746077 - CAT# K32454]

Conferences & Presentations

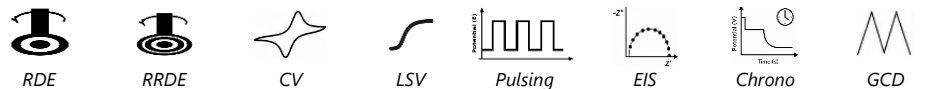
- 2026** 249th ECS Meeting | May, 2026 | Seattle, Washington, United States | Oral Presentation
- 2023** Canadian Chemical Engineering Conference (CSCHE 2023) | October, 2023 | Calgary, Alberta, Canada | Oral Presentation
- 2023** McMaster University Chemical Engineering Conference (MUCEC 2023) | April, 2023 | Hamilton, Canada | Oral Presentation
- 2022** 72nd Canadian Chemical Engineering Conference (CCEC 2022) | Vancouver, British Columbia, Canada | October, 2022 | Oral Presentation
- 2021** McMaster University Chemical Engineering Conference (MUCEC) | March, 2021 | Hamilton, Canada | Host & Moderator
- 2019** The 81st EAGE Conference and Exhibition EAGE 2019 | London, United Kingdom | Oral Presentation
- 2019** American Institute of Chemical Engineers (AIChE) Spring Meeting | New Orleans, Louisiana, United States | March, 2019 | Oral Presentation
- 2016** Qatar Foundation Annual Research Conference ARC'16 | Doha, Qatar | March, 2016 | Poster Presentation
- 2016** 5th Gas Processing Symposium (GPS 2016) | Doha, Qatar | November, 2016 | Oral Presentation
- 2014** American Institute of Chemical Engineers (AIChE) 2014 Annual Student Conference | November, 2014 | Atlanta, Georgia, United States | ChemE Car Competition & Poster Presentation

Honors, Awards & Certificates

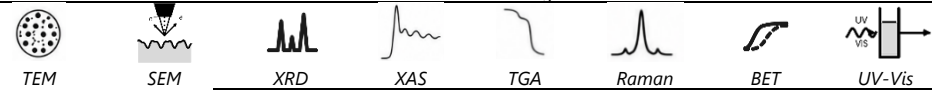
- 2025** Outstanding Teaching Assistant Award | McMaster Engineering Society (MES)
- 2023** Student Travel Award | Canadian Light Source
- 2023** Teaching and Learning Scholar | Paul R. MacPherson Institute for Leadership, Innovation and Excellence in Teaching
- 2022** EGS Travel Award | The Engineering Graduate Society, McMaster University
- 2021** Teaching and Learning Foundations Certificate | Paul R. MacPherson Institute for Leadership, Innovation and Excellence in Teaching
- 2019 – Present** CIRT Associate, Center for integration of Research, Teaching and Learning | Texas A&M University
- 2018** Itochu Fellowship – Chemical Engineering | Texas A&M University at Qatar
- 2017-2018** Texas A&M University "Aggie" Ambassador
- 2013-Present** Omega Chi Epsilon Chemical Engineering Honor Society
- 2012-Present** American Institute of Chemical Engineers (AIChE)
- 2013** Distinguished Student | Texas A&M University at Qatar
- 2012** Dean's Honor Roll | Texas A&M University at Qatar

Summary of Skills

Electrochemical techniques



Material characterization



Coding & automation



Data & image analysis



Crystallography & structure visualization



Process design and P&ID



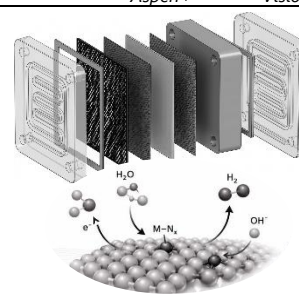
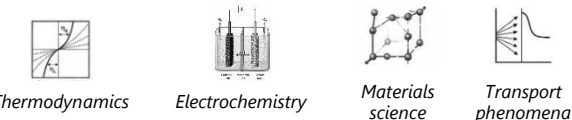
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Citation management



Theoretical knowledge



Professional Affiliations and Leadership Activities

2022-2023	VP Communications, Chemical Engineering Graduate Student Society (ChEGSS) McMaster University
2020-2021	VP for McMaster University Chemical Engineering Conference, Chemical Engineering Graduate Student Society (ChEGSS) McMaster University
2020-2021	Graduate Mentor, Black Student Mentorship Program (BSMP) McMaster University
2018	Executive Education Certifications HEC Paris
2016	Professional Development Summer Institute Teach for Qatar
2014-2015	Stopping Mechanism Team Leader, AIChE ChemE-Car Team Texas A&M University at Qatar

References

Academic

Dr. Drew Higgins

Associate Professor, Chemical Engineering
McMaster University
Email: higgid2@mcmaster.ca

Dr. Nayef Alyafei

Associate Vice President for Technology & Information Security, Qatar University
Former Associate Professor, Petroleum Engineering, Texas A&M University at Qatar
Email: n.alyafei@qu.edu.qa

Dr. Nimir Elbashir

Professor, Chemical Engineering
Hamad Bin Khalifa University
Email: nelbashir@hbku.edu.qa

Professional

Dr. Christine Gabardo

Chief Technical Officer
CERT Systems Inc.
Email: christine@co2cert.com

Dr. Brandon Sutherland

Director, Research Operations
Acceleration Consortium
University of Toronto
Email: brandon.sutherland@utoronto.ca

Dr. Amitava Sarkar

Associate Vice President, Operations & Maintenance
Adani Green Energy/TotalEnergies
Email: amitava.sarkar@totalenergies.com

Teaching

Dr. Jake Nease

Associate Professor and
Department Chair
Chemical Engineering
McMaster University
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Dr. Stuart Linley

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McMaster University
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